TD 3 - Agile Methods

Project Step 1: Plan Mahamma Premium with Scrum
Solutions to exercises T-1 to T-4 (Part B)
National School of Artificial Intelligence - Software Engineering

Sources used: the client request of Ms. Lina Haddad (4 October 2026) and the kick-off meeting notes (12 October 2026). Facts quoted from them: 12,000 free users, about 800 connected at peak, 300 duplication complaints, six developers, eight two-week Sprints, first release in March, nothing signed yet with payment providers, default quiet hours 22:00-07:00, company working week Sunday to Thursday. Everything else is a **team decision or assumption** and is marked as such.

T-1: Vision and Product Goal

(a) Product vision

For individuals, families and small teams (associations, shops, small companies) in Algeria, Tunisia, France and Canada **who** share task lists and cannot afford to lose work or be woken by a badly timed reminder, **Mahamma Premium is** a paid shared to-do service **that** keeps every edit, even made offline, and sends reminders only when the user wants them. **Unlike** today's free version, where one of two offline edits can be lost, lists can be duplicated and reminders ring at 3 a.m., it **never loses a change**, respects each user's quiet hours wherever they are, and is run as a reliable service people can depend on every day.

(b) Product Goal of the March release

Product Goal: by March, an individual or a family can subscribe to Mahamma Premium by card and use shared lists that stay correct (no lost or duplicated change, even offline), with reminders that never ring during their quiet hours and with control over their own data (export and account deletion).

Why it cannot be "everything in the request":

- **Capacity.** Eight Sprints at a forecast of 13 points make about 104 points. The request contains at least ten large features (sync, reminders, payments, teams, export and deletion, support console, three languages, high availability, weak-network performance, AI), which clearly exceed this capacity. The velocity is not even known yet.
- **Dependencies outside the team.** The card contract is only being negotiated and the Algerian gateway (CIB, Edahabia) is not signed. Work depending on them is uncertain.
- **The client herself marks the AI due dates as "later"**, and the team accounts are not yet defined (billing, roles).
- **A Product Goal is one focused objective**, a long-term target the Scrum Team commits to. A goal that means "everything" gives no way to say no, no order of value, and a high risk of delivering nothing that is Done in March.
- The most painful problems of current users (lost work, night reminders) and the revenue objective are covered by the goal. Teams, support console and AI stay in the backlog, ordered after it, and the goal can be revised after a Review.

(c) Stakeholders invited to the Sprint Reviews

Stakeholder	Why they must attend
Ms. Haddad (Product Owner / Product Director)	Decides priorities and accepts the Increment; she also needs to see what marketing can announce before the Ramadan campaign.
Mr. Ouali and the two support agents	They know the real user problems (lost lists, refunds, removed employees) and will operate the support tools; they see if the product reduces their email load.
Representative users: a family organizer, a small-team manager, a free user	Only real use reveals problems, as in the shared-list example of the lecture; feedback on sync and reminders comes directly from them.

Stakeholder	Why they must attend
Legal adviser (when relevant items are shown)	Checks law 18-07 and GDPR items: export, deletion, consent, data location.
Payment contacts (provider / gateway), when payment is shown	Confirm that the integration matches the contract and the refund and renewal rules.
Hosting / operations person of Atlas	Availability and deployment constraints ("almost never down").

T-2: The Scrum Team and Its Rhythm

(a) Who should be the Product Owner?

Option	Advantage	Risk
Ms. Haddad herself	Has the business authority and knows the market, the budget and the campaign date; she is available almost every day, so decisions are quick.	She also runs marketing: limited time for the backlog and a temptation to push urgent marketing requests (like "dark mode today") into the Sprint.
A member of our team	Always present, understands technical effort, and has time to refine the backlog daily.	No real authority over price, priorities or the campaign; becomes a messenger whose decisions the client can overturn. A developer who is also the PO is also overloaded.
A client available only 2 hours every 2 weeks	Gives the strategic view and keeps the PO role with the business.	Questions wait up to two weeks, the backlog becomes stale, and the team builds on guesses. Not enough for a PO, only for the Review.

Decision: Ms. Haddad is the Product Owner. She is accountable for the value of the product and for the order of the Product Backlog (one person, as Scrum requires). To protect her limited time, we agree on: a fixed 30-minute daily slot with the team, a weekly one-hour backlog refinement, and a written rule that new requests (including marketing's) enter the Product Backlog, never the running Sprint Backlog. A team member helps her prepare and write items, but does not decide the order. If she cannot respect the slot, we raise it as an impediment at the Retrospective and propose a delegate chosen by her.

(b) Sprint length and calendar

Sprint length: two weeks (fixed, ten working days, Sunday to Thursday), eight Sprints before March. It is short enough to adapt to the client's changes and long enough to finish a Done item including tests.

Event	When	Duration
Sprint Planning	Sunday of week 1, 09:00	up to 4 hours (maximum for a 2-week Sprint)
Daily Scrum	Every working day, 09:30	15 minutes
Backlog refinement	Tuesday of weeks 1 and 2, afternoon	1 hour (less than 10% of capacity)
Sprint Review	Thursday of week 2, 10:00	up to 2 hours, with stakeholders
Retrospective	Thursday of week 2, 14:00 (after the Review)	up to 1.5 hours, team only

The next Sprint starts on the following Sunday; Friday and Saturday are the weekend, so no event is placed there.

(c) Scrum Master and Daily Scrum

Scrum Master: one team member (named by the group; for example the member with the most Scrum experience in PA/TD), rotating every two Sprints so that everyone learns the role. The Scrum Master keeps the events in their timebox, coaches the team and the PO, and removes impediments (access to production logs, slow code reviews, unavailable client). He or she **does not assign tasks**: the Developers own the Sprint Backlog.

Daily Scrum: 15 minutes, every working day at 09:30, same place (or the same video call), for the Developers only. We inspect progress toward the Sprint Goal in front of the board and the burn-down chart: what helps or blocks the Sprint Goal, what we do today. Impediments are named, then solved afterwards by the people concerned. The PO and stakeholders may listen but do not ask for hours worked or change the Sprint Backlog. A member who is absent posts a short written update in the team chat before 09:30.

T-3: The Initial Product Backlog

(a) Epics of Mahamma Premium

Epic	Source in the request	Main content
E1 Sync and offline editing	Sec. 2, 5, 6	Multi-user and offline edits that are never lost; fix the list duplication bug; instant sync.
E2 Reminders and quiet hours	Sec. 2, 5; notes 7	Reminders before due time; quiet hours 22:00-07:00 (adjustable); time zone when traveling.
E3 Subscription and payment	Sec. 3, 5; notes 4	Monthly or yearly plan; international card; CIB and Edahabia; renewals and refunds.
E4 Teams and sharing	Sec. 4, 5; notes 6	Lists shared with a whole team; roles; removing a member who left.
E5 Privacy and data rights	Sec. 5, 6; notes 5	Export of all lists; account deletion; law 18-07 and GDPR; security.
E6 Support console	Sec. 2, 4, 5	Agents see an account, subscription and payments; refunds; fix accounts.
E7 Reliability and performance	Sec. 2, 6	Service almost never down; works on weak mobile networks; monitoring.
E8 Languages	Sec. 6	Arabic (right to left), French and English in the whole product.
E9 Smart due dates (later)	Sec. 5	AI suggestion of due dates; starts with a spike; not in the March goal.

(b) to (e) Product Backlog (items 1-3 are the given models)

Estimates by planning poker (1, 2, 3, 5, 8, 13), relative to **item 3 = 8 points** (a payment integration with an external provider and subscription states). Points measure effort, complexity and uncertainty, not hours. The reasons for the estimates are given below the table.

#	Epic	Product Backlog item	Pts	Ready? / question for the client
1	E1 Sync	Spike: compare two ways to merge offline edits of the same list (last write per field vs. a merge algorithm) on 100 recorded conflicts	3	Ready (timeboxed: 3 days)
2	E1 Sync	Bug: a shared list is sometimes duplicated after offline synchronization (300 complaints)	3	Ready
3	E3 Payment	As a user, I want to pay monthly or yearly by card, so that I get premium features	8	**Not ready:** which card provider, when signed?
4	E1 Sync	As a user editing a shared list offline, I want my changes and those of others to be merged when I reconnect, so that no change is ever lost	13	**Not ready:** depends on the result of spike 1. Question: if two people change the same field of the same task, is it acceptable to keep the latest and show a notice, or must both be kept?
5	E2 Reminders	As a user, I want no reminder to be sent during my quiet hours (default 22:00-07:00, adjustable), so that I am never woken up at night	3	Ready (rule is clear; the time zone is handled in item 6)
6	E2 Reminders	As a traveling user, I want my quiet hours to follow the right time zone, so that I do not get a reminder at 3 a.m. when I change country	5	**Not ready:** which time zone applies when the user travels: the home one, or the one of the phone (our proposal)? Can the user pin one?
7	E3 Payment	Spike: build a payment interface and test it against the card provider's sandbox; list what the local gateway (CIB, Edahabia) requires for recurring payment, refunds and notifications	3	Ready (timeboxed: 2 days, uses public documentation)
8	E4 Teams	As a team manager, I want to share a list with my whole team, so that every member sees it and new members get access	8	**Not ready:** is a team plan billed per member or flat? Who is the manager? Maximum team size?

#	Epic	Product Backlog item	Pts	Ready? / question for the client
9	E5 Privacy	As a user, I want to export all my lists and delete my account and data, so that I control my personal data	5	**Not ready:** question for the legal adviser: export format, retention period, and what happens to a shared list when its owner deletes the account?
10	E6 Support	As a support agent, I want to find a user's account and see subscription status and payments (read only), so that I can answer without long email exchanges	5	**Not ready:** Mr. Ouali: which data may agents see (privacy), and who may refund?

Why these estimates (planning poker):

- **Item 4 (13):** the heart of the product promise; algorithm, tests on two devices at the same time and offline, high uncertainty. 13 is too large for one Sprint, so it must be **split** after spike 1 (done in T-4).
- **Item 5 (3):** small, well-defined rule, easy to test at the boundaries (22:00, 07:00, midnight).
- **Item 6 (5):** more than item 5: detecting the time zone and handling changes, with an open rule.
- **Items 8 (8) and 3 (8):** both create a new concept (team with roles, subscription) with external or legal rules, so they are about the same size as the reference.
- **Items 9 and 10 (5 each):** clear scope, but legal and privacy questions; smaller than the reference because there is no external provider.

(d) Order of the backlog, and the first five items

Rank	Item (#)	Pts	Reason for its place
1	#1 Spike: merge strategy	3	Biggest risk and the largest item (#4) depend on it. It is a short spike, done first so the team learns before building.
2	#2 Bug: duplicated lists	3	Ready, 300 complaints, destroys trust. It may share its root cause with the merge problem, which the spike will show.
3	#5 Quiet hours	3	Ready, cheap, and removes the 3 a.m. complaint. Can be built test first (P-1) with no external dependency.
4	#4 Merge offline edits	13	The central promise: "changes must never be lost". Comes right after the spike; it will be split to fit a Sprint.
5	#7 Spike: payment	3	Nothing is signed yet: learning early about the providers reduces the largest external risk before item #3, and the revenue objective depends on it.
6	#3 Pay by card	8	Revenue, but not ready until the contract and spike are done.
7	#6 Time zone when traveling	5	Waits for the client's answer.
8	#9 Export and delete	5	Legal obligation (law 18-07, GDPR) before selling in Europe.
9	#8 Team lists	8	High demand, but outside the March Product Goal if capacity is short.
10	#10 Support console (read only)	5	Reduces support overload; can start with the read-only view.

The order follows value for the user and the business, risk reduction (spikes early) and dependencies (merge strategy before merge feature; payment learning before payment). The total of the ten items is 56 points; the remaining backlog items (see below) will be added in the 15-item version of A-1.

Candidates for items 11 to 15 (A-1): remove a member who left a team (3); subscription states: trial, active, canceled, expired (5); refunds from the support console (5); Arabic right-to-left interface and French/English check (8); monitoring and automatic failover for high availability (8); faster sync on weak networks (5); smart due dates spike (3, later).

T-4: Sprint 1 and the Definition of Done

(a) Sprint Goal and selected items

> **Sprint Goal (Sprint 1):** shared lists are no longer duplicated, two devices that edit different tasks offline keep both changes, and no reminder is sent during quiet hours; the team chooses its merge rule on evidence.

Item 4 (13 points) does not fit with the others, so it is **split** into a slice that gives value now and a remainder:

- **4a (4 pts):** when two devices edit *different* tasks of the same list offline, both changes are kept after reconnection. It works with either merge approach, so it does not wait for the result of the spike.
- **4b (8 pts):** same-field conflicts, notices to the user (stays in the backlog, rank 4, after spike 1).

Item	Pts	How the Developers will do it (Sprint Backlog, tasks)
#1 Spike: merge strategy	3	Replay the 100 recorded conflicts through both approaches; count lost edits; write a decision record; present at the Review (first 3 days).
#2 Bug: duplicated lists	3	Reproduce with production-like data; find the cause (retry creating a list during sync); fix it with a unique id; regression test; check the 300 complaint cases.
#5 Quiet hours	3	Write failing tests first (inside, outside, boundaries, crossing midnight); implement is_quiet; connect it to the reminder sender; setting for start and end.
#4a Merge, different tasks	4	Two-device (simulated clients) test, merge rule for different tasks, API contract update, test for edits made at the same time.
Total	**13**	Equals the forecast. Item #7 (3 points) is the first one pulled if the team finishes early.

The Sprint Goal is one coherent objective ("users can trust their lists and their nights"); it gives the team flexibility on how to reach it. The PO agrees that nothing else (for example a new marketing request) enters the Sprint without a discussion with the Developers, and that the Review shows the Increment to real users.

(b) Definition of Done

An item is **Done** only if all the following points are true. Each point is checkable by someone else.

#	Definition of Done item	General / Mahamma
1	Code reviewed and approved by at least one other developer, then merged into the main branch.	General
2	Automated tests (pytest) cover every acceptance criterion; the continuous integration pipeline is green, with all earlier tests still passing.	General
3	Linter and type checks pass; no known critical or major bug remains on the item.	General
4	Deployed on the staging environment, and the acceptance criteria are shown to the Product Owner (it could be released).	General
5	The API contract and the user documentation are updated when behavior changes; any technical decision is recorded in a decision record.	General
6	Any database change comes with a migration tested on a copy of production-like data.	General
7	**Sync:** tested with two devices (or two simulated clients) editing the same list at the same time and offline: zero lost edit and zero duplicated list.	**Mahamma**
8	**Reminders:** tested at the quiet-hours boundaries (exactly 22:00 and 07:00, across midnight) and with at least two time zones.	**Mahamma**
9	**Languages:** every new text exists in Arabic, French and English, and the Arabic screen is checked right to left.	**Mahamma**
10	**Privacy:** no personal data in logs, and every new personal data field is added to the data inventory (law 18-07 and GDPR).	**Mahamma**

The Definition of Done is the team's commitment for the Increment; the Retrospective may add items (for example after a bug found in the Review).

Exit ticket (individual)

First item in the Product Backlog: item 1, the merge-strategy spike. It is first because the central promise ("changes must never be lost") and the largest item (#4, 13 points) depend on it, it is cheap (3 points, timeboxed to 3 days) and it reduces the biggest technical uncertainty before the team builds on it.